



(EN) PRODUCT NAME

Human Immunodeficiency Virus HIV 1/2 . A rapid, one-step assay for the qualitative detection of antibodies to human immunodeficiency virus HIV-1/HIV-2 in human serum, plasma and whole blood. For in vitro diagnostic use only.

PURPOSE

The reagent is used for qualitative detection of HIV 1/2 antibody in serum/plasma/whole blood. Human immunodeficiency virus is the causative agent of AIDS (acquired immunodeficiency syndrome). AIDS is the final stage of a long process in which an infected person's immune system and its ability to control infections or malignant proliferative disorders is progressively destroyed and is transmitted mainly through sexual contact in general. Also mother-to-child blood transmission through transmissible infection in three ways.

PRINCIPLE

The test uses antibodies including mixed HIV1 and HIV2 antigen (P24, gp120, gp41, gp36) and rabbit anti-HIV antibody on colloidal gold nitrocellulose membrane labeled with HIV antigen as a tracer. The reagent is used for detection of HIV 1/2 antibody in serum/plasma/whole blood according to the principle of double antigen sandwich method and gold immunochromatographic assay.

MAIN COMPONENTS

Main components: sample pad, colloidal gold pad, nitrocellulose membrane, absorbent paper and PVC board. Colloidal gold labelled substrate coated with mixed HIV antigen (P24, gp120, gp41, gp36), nitrocellulose membrane coated with mixed HIV antigen (P24, gp120, gp41, gp36), control line coated with rabbit anti-HIV polyclonal antibody.

STORAGE AND SHELF LIFE

Store packed in a sealed bag at 4-30°C, avoid hot and sunlight, in a dry place, valid for 24 months. DO NOT FREEZE! Some protective measures should be taken during hot summer and cold winter to avoid high temperatures or freeze-thaw.

SAMPLE REQUIREMENTS

Use fresh whole blood samples. Whole blood samples with anticoagulant can be stored at 2-8°C for 24 hours and should be used immediately without anticoagulant. Frozen whole blood samples should not be used. (If bleeding occurs, it is

recommended to use serum samples.)

MATERIALS

1. Test cartridge
2. Lancet
3. Capillary dropper
4. A packet of alcohol swabs
5. Diagnostic solution
6. Leaflet and method of use

Materials needed but not provided:

- Timer

RESEARCH METHOD

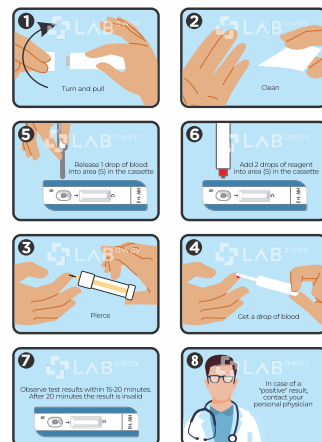
Instructions for use should be read in full before testing. Allow the controls on the test device to equilibrate to room temperature for 30 minutes (20°C-30°C) before testing. Do not open the inner packaging until ready to use, it should be used after one hour if opened.

Tape:

1. Wash your hands with soap and rinse thoroughly with clean warm water.
2. Leave the box at room temperature before opening it.
3. Open the box, remove the test cartridge and place it on a clean and flat surface. Then remove all other accessories from the box and place them in a convenient place near you. Perform the test within one hour, and you will get the best results if the test is performed immediately after opening the foil pouch.
4. Carefully pull and discard the released lancet cap. Use an alcohol swab to clean the fingertip /middle or ring finger/ that will be punctured. Allow to air dry.
5. Press the lancet on the front side where the cap is removed. The nozzle retracts automatically and safely after use. Massage the hand without touching the puncture site, massaging the hand towards the fingertip of the middle or ring finger to be punctured. Holding the hand down, massage the end that is pricked to get a drop of blood. Without squeezing the dropper bulb, place it in contact with the blood. The blood migrates into the capillary dropper through the capillary to the line indicated on the capillary dropper.
6. Using the dropper, vertically transfer two (2) drops (50-60µl) of the whole blood sample onto the sample test portion (S) of the cartridge, avoiding bubbles. Allow the blood to completely disperse in the well, then clean your finger with alcohol and a swab.
7. Unscrew the cap of the reagent bottle and add one (1) drop (40-50ul) of solution to the test area (S) and start the timer.

8.Observe the test results immediately within 15-20 minutes, the result is invalid after 20 minutes.

Do not read the result after 20 minutes. To avoid confusion, discard the test device after interpreting the result.



INTERPRETATION OF RESULTS

- Positive**
Two distinct red lines are displayed. A red line in the control region (C), and another red line in the test region (T). A pink to red line (T), even if very narrow, indicates a positive result.
- Negative**
A red line appears in the control area (C). No line appears in the test area (T).
- Invalid**
No red lines or the control line is not displayed. Insufficient specimen or incorrect procedural techniques are the most likely causes. Review the procedure and repeat the test with a new device. If the problem persists, stop using the kit and contact your local distributor.

RESTRICTIONS

This reagent is designed for a qualitative screening test. HIV 1/2 concentration cannot be determined by this qualitative test. Reagent results are for clinical reference only, which is not the sole basis for clinical diagnosis and treatment. A confirmed diagnosis and treatment should only be made by a physician after evaluation of all

clinical and laboratory results. Positive results from patients who have received blood transfusions or therapy with other blood products should be analyzed carefully. Abnormal results may occur depending on operator error or drug use. If AIDS is still suspected, a sample should be taken later and retested. A negative result does not completely rule out HIV 1/2 infection, it only indicates that HIV specific antibodies are not present or the concentration is below the limit of detection. Mutation, silent infection and sensitivity will also result in negative detection results.

PERFORMANCE CHARACTERISTICS

Negative specificity:

The match rate must exceed 18/20 when detecting national negative quality control samples with HIV 1/2 kits.

Positive specificity:

The matching rate should be 20/20 when detecting national positive quality control samples with HIV 1/2 kits.

Detection limit:

At least one of the three results must be positive on HIV national quality control border sample detection with the HIV 1/2 kits. The match rate must be at least 1/3, the plasma matrix (S1) must give a negative result. The plasma matrix (S1) must give a negative result, while S2 and S3 give a positive result when detecting quality control samples with an internal limit.

Repeatability:

The results should be consistent and the degree of staining should be consistent when detecting national/international precision control samples from 10 sets of the same batch.

Analytical sensitivity:

Bilirubin concentration below 146.7 pmol/L, triglycerides concentration below 8.74 mmol/L had no effect on the detection results. Haemolysis with Hb concentration above 6.5 g/L may affect the detection results. Another detection method is suggested.

The addition of HAV positive samples, HBsAg, HBeAg and HBcAb positive samples, HBsAg, HBeAg and HBcAb positive samples, HCV positive samples, HEV positive samples and syphilis positive samples showed no cross-reactivity. Rheumatoid factors would not affect the results. A hook effect will not occur even if the HIV 1/2 concentration is high.

Diagnostic specificity and sensitivity: A clinical evaluation was performed comparing the results obtained

using the rapid HIV 1/2 antibody test kit (immunochromatography) and the EIA HIV 1/2 antibody test.

The results are as follows:

Method	E I A		Final result
	Results	Positive	Negative
HIV 1/2 Ab test kit	Positive	516	0
	Negative	0	581
Final result		516	581

Relative sensitivity: 100%

Relative specificity: 100%,

Accuracy: 100%.

PRECAUTIONS

The result is invalid after 20 minutes. Do not use after expiry date. The strength of the qualification control line does not indicate a problem with the reagent qualification, a test result that is clearly visible demonstrates that the reagent is effective. During testing, dark and light color may appear, even very light mark. If the blood is too thick and the climb rate is less than 4 mm/min, add one drop of the solution to the sample pad to help. Do not use other types of qualified control samples to test the reagent. Components from different batches cannot be interchanged to avoid erroneous results.

MANUFACTURER

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Phone: +359 882 399 001
See instructions for use .
Disposable.
Store between 4-30 °C



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